SPECIFICATION OF VIBRATION MEASUREMENT

I GENERAL

- This specification outlines the floor vibration field test requirement. It is the intent of this specification to carry out vibration test in accordance to ISO 2631-2:1989 “Mechanical vibration and shock – Evaluation of human exposure to whole-body vibration – Part 2: Vibration in buildings (1 Hz to 80 Hz)” to determine compliance with floor vibration criteria in HKBEAM Plus guideline for the following area:

Location	Criterion Curve	Vibration Velocity, RMS (1 to 200 hz)
Office	Office	400 μm/s, RMS

II OBJECTIVE

- Vibration testing shall aim in characterizing vibration due to different sources, such as E&M machinery. The vibration test is intended to collect the floor vibration data to determine compliance with the Specification and HKBEAM Plus requirement.

III SCOPE OF WORK

- Coordinate for test arrangement and schedule
- The Contractor shall submit a method statement and detail work programme of vibration test method statement to Architect for approval.
- The Contractor shall appoint an accredited independent laboratory to carrying out the floor vibration test.
- The Contractor shall provide all necessary support and coordination to facilitate the vibration test conducted by independent acoustic laboratory.
- During the mechanical machinery vibration testing, equipment shall be operated under design operation load.
- The Contractor shall submit the test report with full records of data, findings and recommendations, comparison of project vibration criteria to the Architect within 21 days after the field measurement.

IV VIBRATION CRITERIA

- The vibration criteria are expressed in the form of 1/3 octave band, constant velocity lines in the range of 1 to 200Hz in accordance to the ASHRAE and ANSI S3.29-1983.
- The vibration criteria for the project, the frequency range spans from 1 to 200 Hertz (Hz):

Location	Criterion Curve	Vibration Velocity, RMS (1 to 200 hz)
Office	Office	400 μm/s, RMS

*Note: Measurement Parameter for mechanical machinery (continuous steady source) shall be Average Vibration Level.

V PERSONNEL

- The Contractor shall appoint an accredited independent laboratory for carrying out the vibration measurement. All the measuring personnel shall have acquired adequate training with a minimum of 10-year experience in field vibration measurement exercise and at least 10 years Post Experience as Hong Kong Institute of Acoustics member. Resume of the proposed person shall be submitted for the Architect's approval.

- Costs for retesting construction that replaces or is necessitated by work that failed to comply with the Contract Documents shall be borne by the Contractor at no cost to Employer.

VI TEST REQUIREMENT

- The Contractor shall carry out all adjustments as are necessary to maintain the E&M equipment in satisfactory by the Architect throughout the Field Measurement Period.

- Check all control functions, from all controlling devices to all controlled devices, for proper operation.

- Adjust, balance and align all equipment for optimum quality and to meet the noise and vibration control specification.

- The Field Measurement will be conducted in the presence of the Architect and/or Project Acoustic Consultant.

- During measurements of ambient testing, all mechanical equipment for buildings including AHUs, chiller plant, cooling towers, genset, pumps and fans must be off.

- Vibration testing may require suspension of ongoing construction activities to prevent or limit data contamination. As such it is important that vibration testing is coordinated closely with the construction management and appropriately scheduled to allow for timely completion.

- The measurement report shall be endorsed by a Corporate Member of Hong Kong Institute of Acoustics (HKIOA) or equivalent to determine whether the specified vibration criteria and the requirement of HKBEAM Plus HWB 7 are achieved.

VII INSTRUMENTATION

- The Contractor shall be responsible for providing test equipment for the field tests.

- The proposed vibration instrument shall be submitted to Architect for approval, including but not limited to the following:

- Vibration Analyzer – data in 1/3 octave band, frequency range spans from 1 to 200Hz.

- The Contractor shall supply test certificates for all instruments and re-calibrate instruments as necessary.

VIII LOCATION FOR VIBRATION MEASUREMENT

- This shall be determined and agreed with the Project Manager, Architect and project acoustic consultant. The proposed testing locations shall be submitted for Project Manager, Architect and the project accountant's approval. The selection of vibration test locations shall include but not limited to the following:

  a) All Office type premises: Security Control Room & Management Offices / Network Operation Centre / Office

  b) All Conference Rooms

  c) All occupied areas are the most sensitive to the vibration excitation from all building services equipment.

IX TEST METHOD AND PROCEDURE

- Test method and procedures should follow the agreed standards defined either by the Project Acoustic Consultant or as laid down in the Specification.

- Vibration readings will consist of measurements multiple locations and different mechanical equipment. Vibration measurement will particularly focus on the contribution of internal mechanical system and ambient condition.

- Record vibration readings on ground and elevated floor levels.

- Measure floor vibration at the centre of the floor bay. Take minimum two readings for each measuring points.

- The accelerometer shall be adhesively attached to parent floor slab.

- Vibration testing consists of mounting accelerometer on to the test surface and recording vibration levels. Low noise, high sensitivity transducer shall be provided to properly record vibration levels on site, particularly at very low frequencies.

- Test data shall be stored on the analyzer that is controlled by computer for record and analysis. All readings shall be in vibration velocity levels (μm/s, RMS) and the data shall be analyzed using 1/3 octave band format.

X QUALITY ASSURANCE

1. Testing Agency Qualifications: A HOKLAS-accredited independent agency with the experience and capability to conduct testing and inspecting indicated, with minimum 5-year equivalent acoustical measurement experience.

   a) HOKLAS-accredited: A testing laboratory accredited under the Hong Kong Laboratory Accreditation Scheme.

2. Reports for Responsible Field Testing: Prepare and submit certified written reports to advise if comply with project requirements and HKBEAM Plus's guideline that include the following. The measurement report shall be endorsed by a Corporate Member of Hong Kong Institute of Acoustics (HKIOA) or equivalent.

   a) Date of issue.

   b) Project title and number.

   c) Name, address, and telephone number of testing agency.

   d) Dates, times and locations of tests.

   e) Equipment list with model name and serial number.

   f) Names of individuals making tests and inspections.

   g) Description of the work and test method.

   h) Identification of products.

   i) Complete test data and chart.

   j) Test results and an interpretation of test results.

   k) Comments or professional opinion on whether tested work complies with the Contract Document requirements, and if non-compliant observed causes and particulars of non-compliance.

   l) Photo record for each sampling points

   m) Certificates of calibration for test equipment.

   n) Name and signature of laboratory inspector.

   o) Recommendations on retesting.

END OF SECTION